

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 1000+060

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – VUP

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

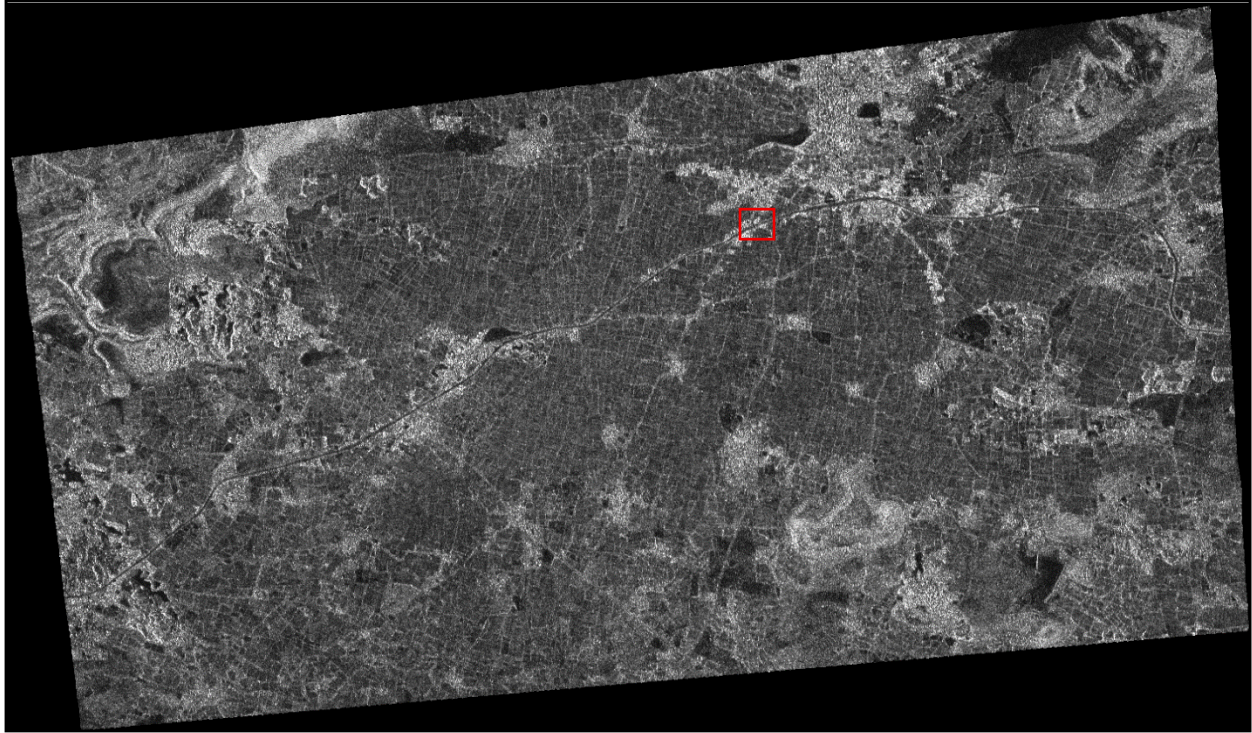
Flight Direction - DESCENDING

Polarization – VV+VH

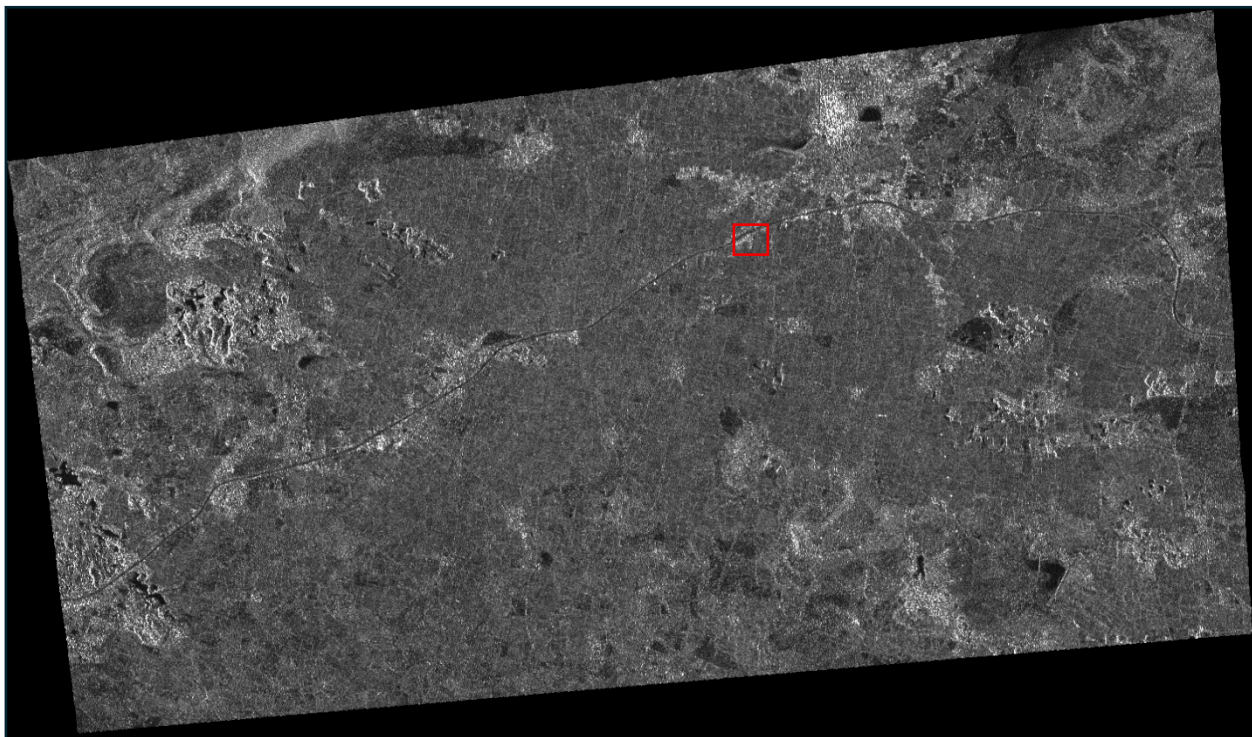
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization



3. Results of SAR based bridge inspection, including visualizations

Bridge Components

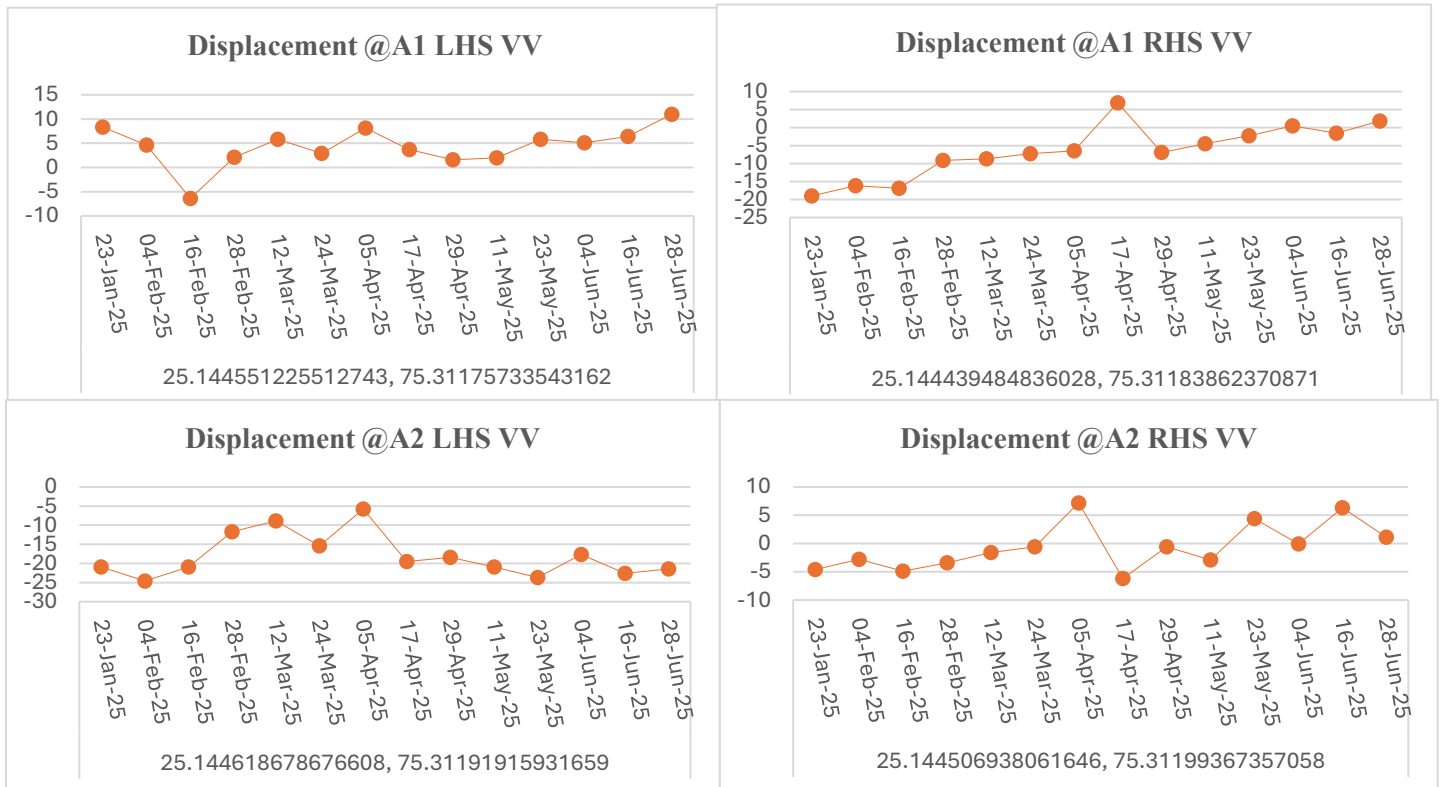
VUP – CH-1000+060

Length – 33.91M

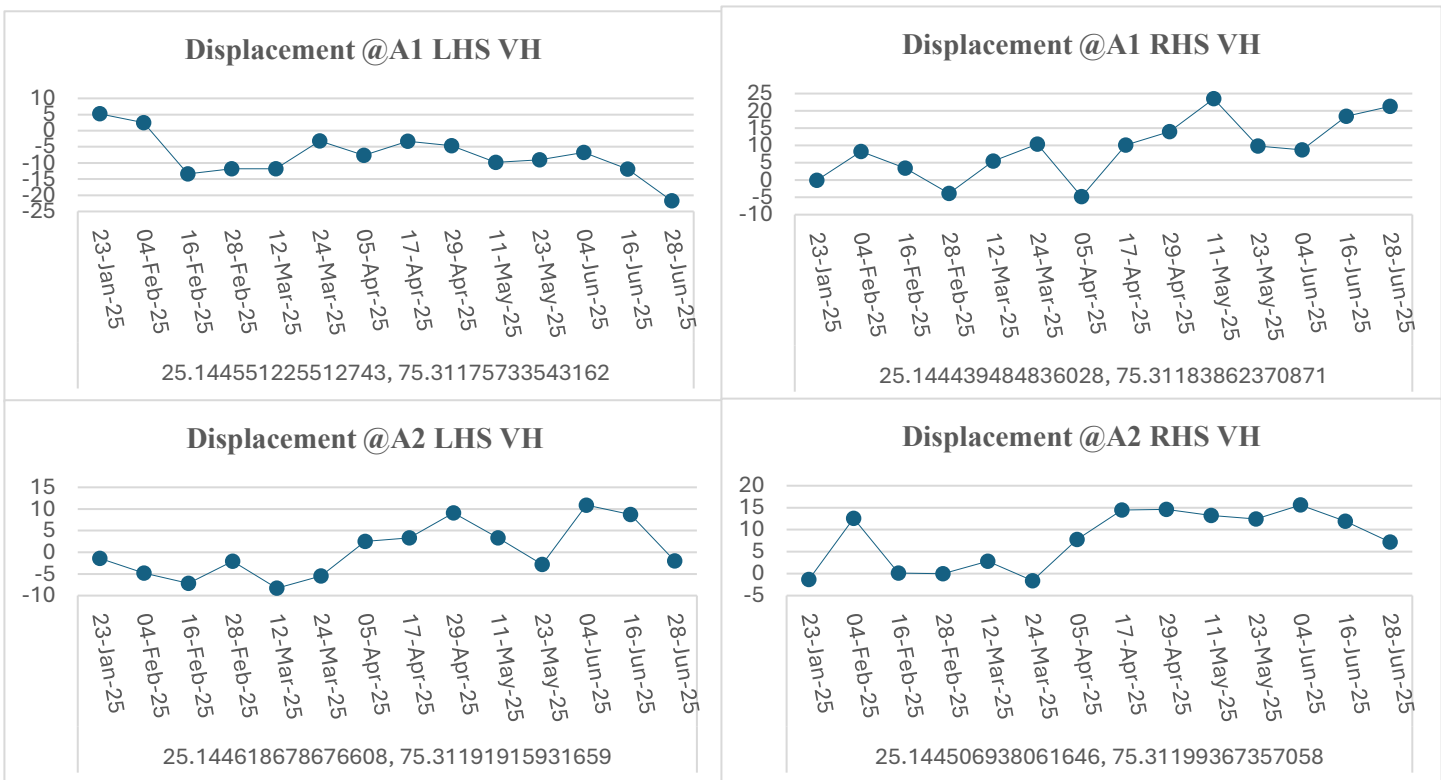


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



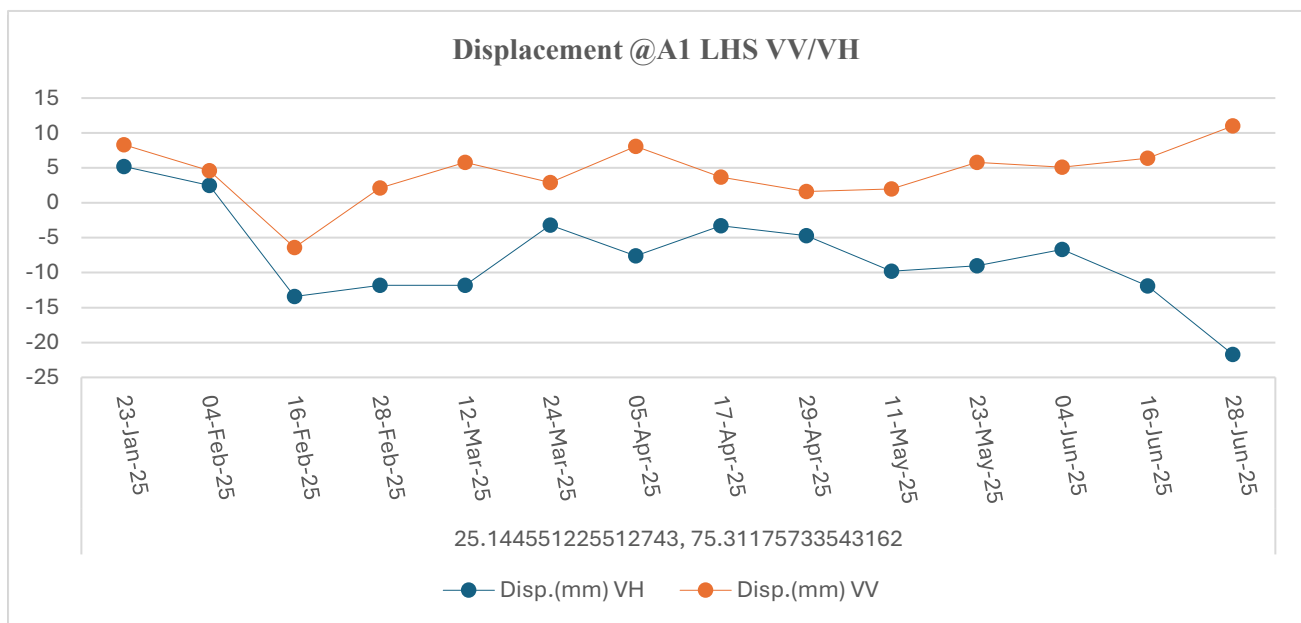
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.144551225512743, 75.31175733543162)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** exhibits alternating uplift and settlement, but with an overall upward (positive) bias.
- **Horizontal Displacement (VH)** Horizontal displacement at A1 LHS shows a predominantly negative trend, indicating sustained lateral movement in one direction.

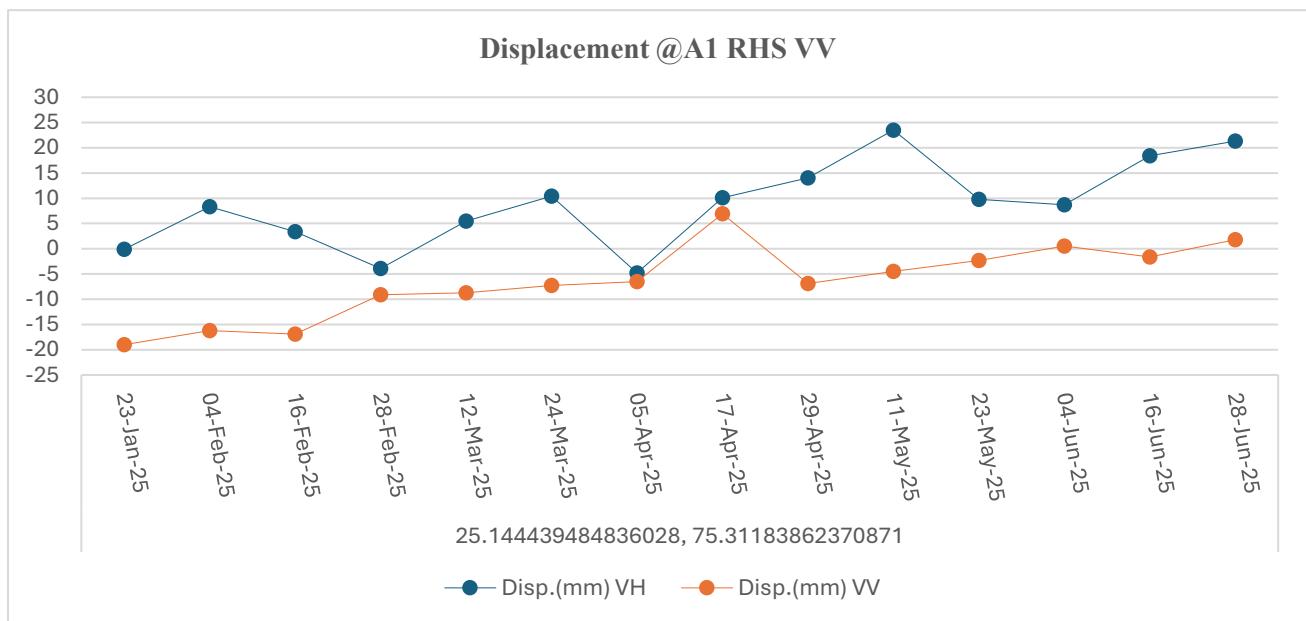
Critical Events & Anomalies:

- **January 2025:** VH begins near +5 mm and rapidly shifts to -14 mm by mid-February, while VV transitions from +8 mm to -7 mm, indicating an early lateral shift accompanied by short-term settlement.
- **June 2025:** VH reaches its maximum negative displacement (-22 mm) by late June, while VV rises to approximately +11 mm, highlighting strong lateral movement dominance with simultaneous vertical uplift, which may indicate rotational or shear-type deck response at A1 LHS

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.144439484836028, 75.31183862370871)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** at A1 RHS shows a strong positive trend with significant variability, indicating dominant lateral movement.
- **Vertical displacement (VV)** remains consistently negative for most of the period, indicating sustained settlement.

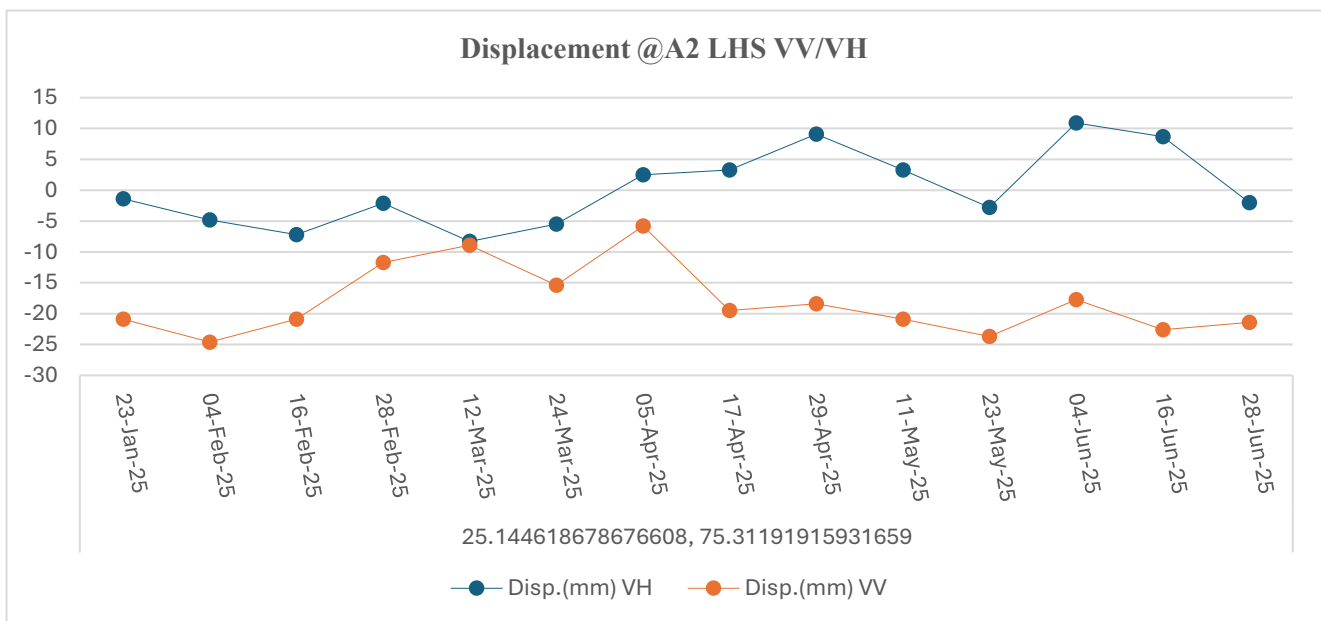
Critical Events & Anomalies:

- **January-February 2025:** VH starts near 0 mm and rises to approximately +7 mm, while VV remains between -18 mm and -15 mm.
- **March 2025:** VH fluctuates between +4 mm and +10 mm, while VV remains negative suggesting continued settlement with moderate horizontal drift.
- **May-June 2025:** VH increases sharply, reaching maximum values of +23 mm, while VV stays mildly negative. This divergence indicates dominant lateral displacement with persistent settlement, suggesting possible torsional behaviour or stiffness imbalance at the RHS support.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.144618678676608, 75.31191915931659)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** at A2 LHS exhibits moderate variability with a clear positive shift during mid to late monitoring.
- **Vertical Displacement (VV)** Vertical displacement shows a consistently negative trend, indicating dominant settlement behaviour throughout the monitoring period.

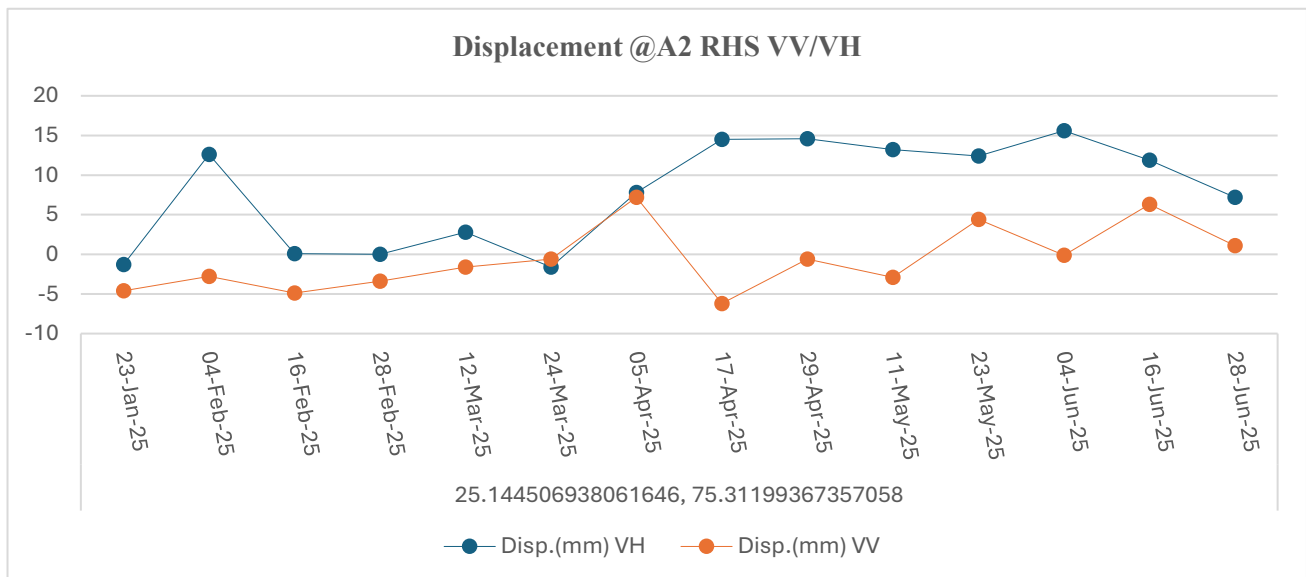
Critical Events & Anomalies:

- **January-February 2025:** VH begins between -2 mm and -7 mm, while VV remains strongly negative, indicating initial lateral stability combined with significant vertical settlement.
- **June 2025:** VH fluctuates between +11 mm and -3 mm, while VV remains consistently negative, confirming persistent settlement with intermittent lateral oscillations toward the end of the monitoring period.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.144506938061646, 75.31199367357058)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacement exhibits low-to-moderate variability, alternating between minor settlement and uplift.
- **Horizontal (VH) displacement** at A2 RHS shows a clear increasing trend after March 2025, indicating progressive lateral movement.

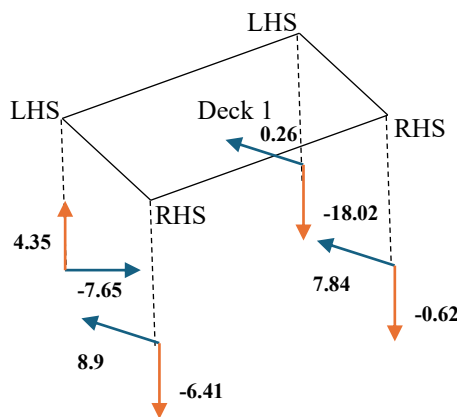
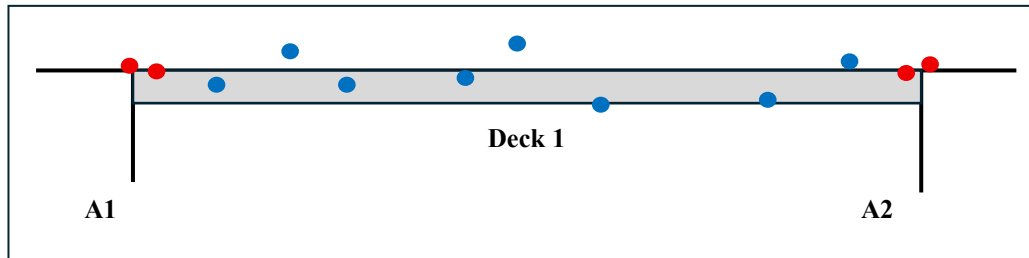
Critical Events & Anomalies:

- **January-February 2025:** VH begins slightly negative and spikes sharply to +12 mm in early February, while VV remains mildly negative, indicating early lateral instability with limited vertical response.
- **March 2025:** VH drops back near 0 mm, while VV gradually improves toward -2 mm, suggesting a temporary lateral stabilization phase before major movement onset.
- **May-June 2025:** VH remains consistently positive before gradually reducing to 7 mm by late June. VV oscillates between -3 mm and +6 mm, confirming persistent lateral displacement with secondary vertical fluctuations toward the end of monitoring.

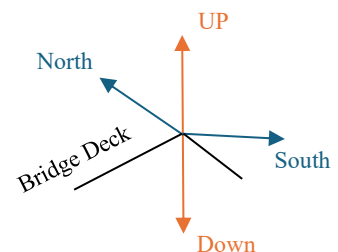
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- The bridge deck slab exhibits clearly non-uniform vertical displacement, indicating differential vertical response between the LHS and RHS regions. The coexistence of uplift on the LHS and significant settlement on the RHS confirms strong vertical rotation combined with localized bending, rather than uniform vertical deflection.

Horizontal Displacement Behaviour:

- Horizontal displacement demonstrates pronounced directional variability across the deck width, reflecting torsional deformation behaviour. The opposing displacement directions and uneven magnitudes confirm torsional rotation about a longitudinal axis, rather than rigid-body translation

CONCLUSION:

- The bridge deck slab demonstrates a torsion-dominated deformation response, characterized by severe vertical settlement on the RHS combined with asymmetric uplift and lateral displacement. Overall, the magnitude and asymmetry of the observed displacements indicate structurally significant behaviour, warranting continued monitoring and detailed assessment, particularly with respect to torsional effects and support performance.